

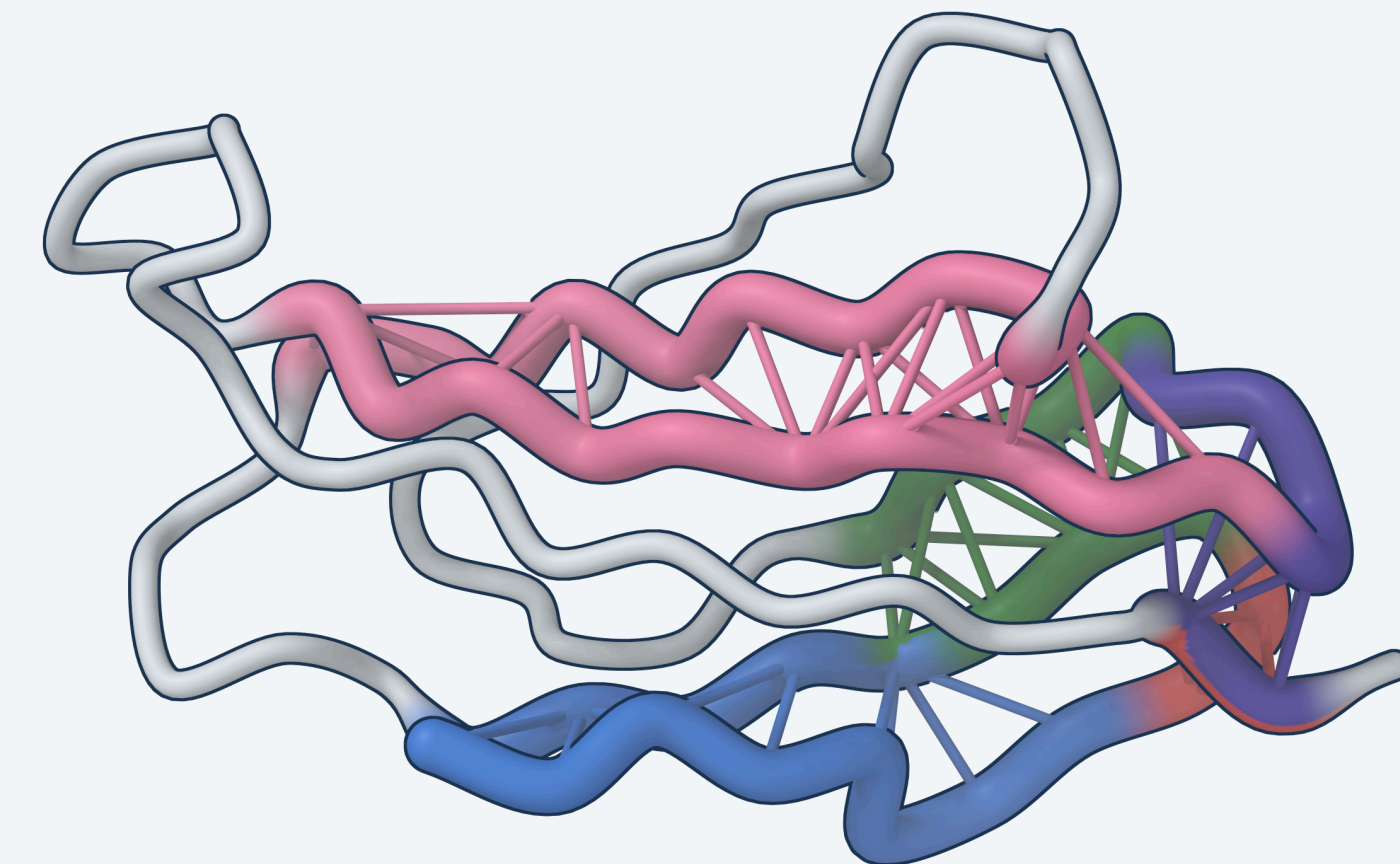
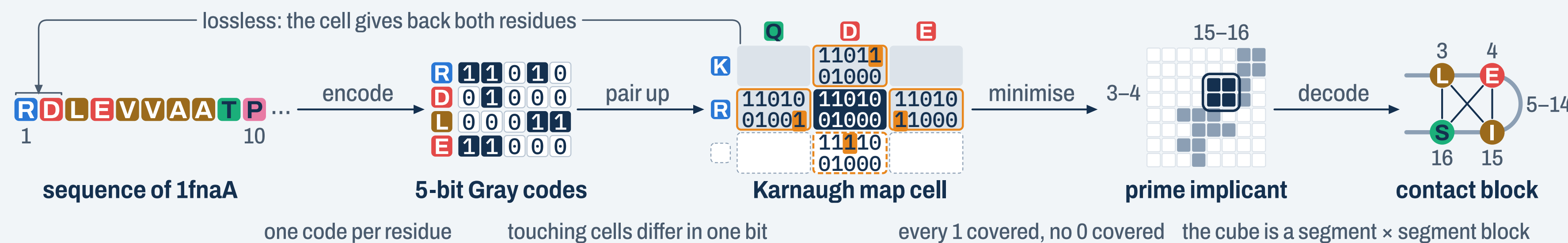
Is KMAP Possible? Formal Verification of Karnaugh Map Encoding for Biological Sequences in Lean 4

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000 The finding*

We formalised in Lean 4 the mathematics that turns DNA, RNA and protein sequences into Karnaugh maps: 131 machine-checked theorems, none left unproved. On this foundation every structure becomes an exact Boolean circuit with exception-free rules. So does a viral species, given a dataset of its sequences: here 1,299 SARS-CoV-2 Omicron Spike proteins.

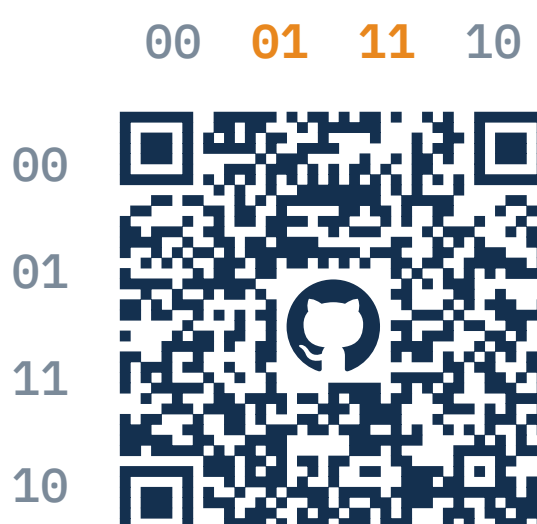
Gray-coded grids of sequences, from Petoukhov's genetic Karnaugh matrices¹ to chaos-game maps (FCGR)^{2,3}, are usually taken on trust. Here the encodings are theorems and every circuit and rule is checked on every case.



Strand pairs, found as blocks

- 25-34 with 61-70: 6 blocks
- 1-8 with 14-18: 4 blocks
- 11-14 with 53-56: 3 blocks
- 57-60 with 89-90: 2 blocks
- 9-10 with 89-90: 1 block

Turn any protein into a circuit



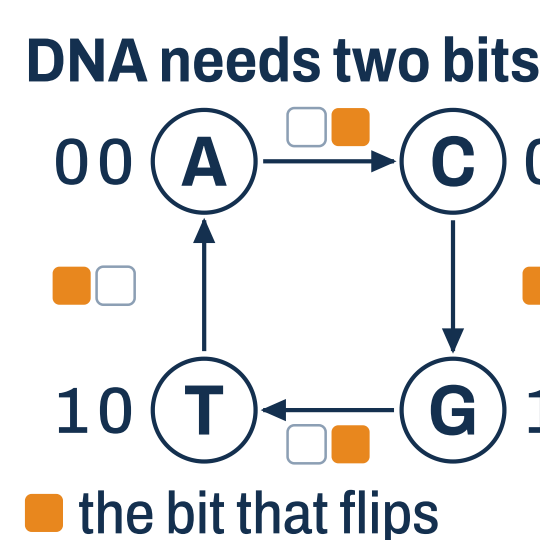
Any PDB entry, step by step, to its rules and inferences.

Fibronectin type-III domain (PSICOV 1fnaA, 91 residues). Its exact contact circuit, read as rules: block rules that continue one another are strand pairs, one colour each.

001 A code that changes one bit at a time

Binary counting can flip many bits at once: 0111 $\rightarrow$ 1000 flips four. The reflected Gray code $g(n) = n \oplus (n \gg 1)$ visits every number once and changes exactly one bit per step. DNA needs two bits, and A C G T is itself a Gray cycle.

n	binary	flips	Gray $g(n)$	flips
0	000		000	
1	001	1	001	1
2	010	2	011	1
3	011	1	010	1
4	100	3	110	1
5	101	1	111	1
6	110	2	101	1
7	111	1	100	1

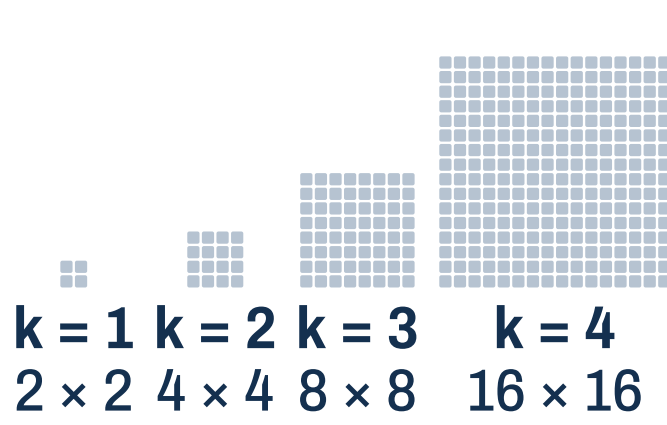


011 A Karnaugh map⁴ is a folded truth table

Order a grid's rows and columns by the Gray code and touching cells differ in one variable, even across the edges, which wrap like a torus. Put a k -mer's first letters on the rows and the rest on the columns: each 2-mer owns one cell of a 4×4 map, each 4-mer one of a 16×16 map.

first base	second base			
	A 00	C 01	G 11	T 10
A 00	AA	AC	AG	AT 0 1 0
C 01	CA 0 1 0	CC	CG 0 1 1	CT 0 1 1
G 11	GA	GC	GG	GT 1 1 0
T 10	TA	TC	TG	TT

One cell per k -mer
4 $\times$ the cells per base



010 Amino acids on a five-dimensional cube

Proteins need five bits: 32 codewords for 20 residues. Numbering residues group by group before Gray-coding keeps chemistry close. All $20 \times 19 / 2 = 190$ pairwise distances are proved: 40 pairs sit one bit apart, only four sit five apart, and the code uses exactly half of the cube's 80 edges.

Flattened into a Karnaugh map: touching cells differ in one bit, and the edges wrap

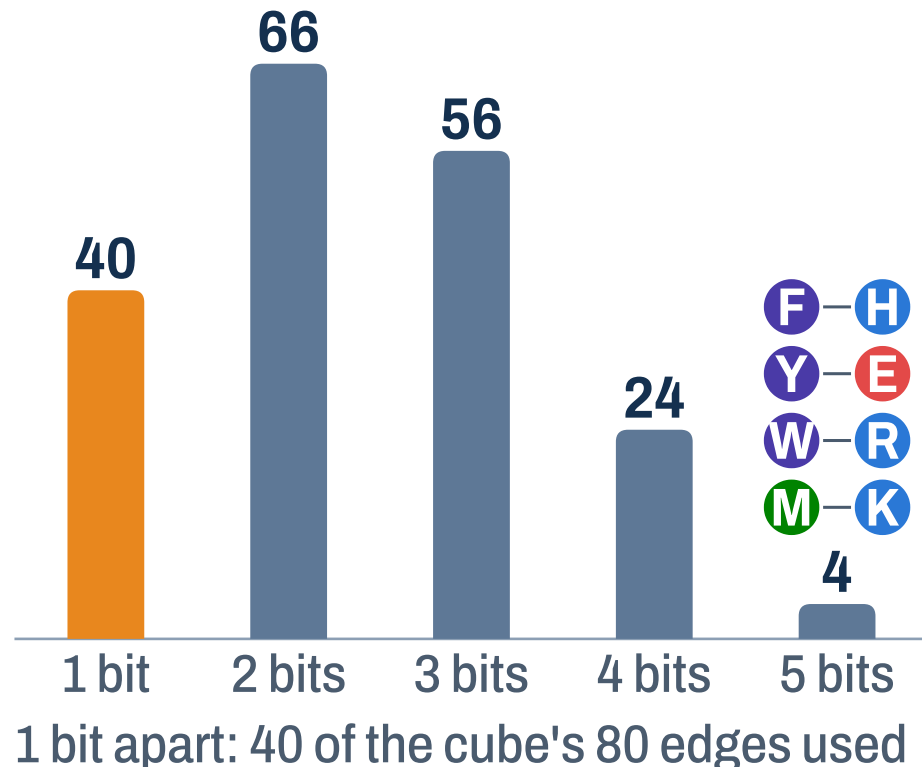
first 2 bits: 000 001 011 010 110 111 101 100

last 3 bits: 000 001 011 010 110 111 101 100

00	A	V	L	I	F	Y	W	M
01	D	Q	N	T	S	G	P	C
11	E	H	K	R				
10								

- aliphatic hydrophobic
- aromatic
- sulfur-containing
- structure-breaking
- polar uncharged
- negatively charged
- positively charged
- unused codeword (12)

How many bits apart: all 190 pairs



110 Contacts as a Boolean function

Contact map (Boolean function): $C(i,j) = 1 \Leftrightarrow C\beta - C\gamma < 8 \text{ \AA}, |i - j| \geq 6$.

- Quine-McCluskey⁵: merge cells one bit apart.
- Prime implicant: a cube that cannot grow.
- Circuit: OR of covering primes = the map.
- Lean, 8 residues: symmetric, no merges.

Merge cells one bit apart

AB	00	01	11	10	CD
00					
01		1	1		
11		1	1		
10		1	1		

8 residues, 4 contacts

0101+0111 $\rightarrow$ 01-1	000				
1101+1111 $\rightarrow$ 11-1	001				
01-1+11-1 $\rightarrow$ -1-1	011				
1000+1001 $\rightarrow$ 100-	010				
	110				
	111				
	101				
	100				

$f = BD + ABC$
dashed: prime, not needed

no two touch, so each is its own prime implicant

111 How Lean checks a claim

- Lean 4⁶: statement = type $\cdot$ proof = program $\cdot$ a small kernel checks it.
- native_decide = compiled exhaustive check: runs every case ($\leq 2^{20}$ per theorem), trusting the compiler.
- 0 sorry $\cdot$ 0 axioms of our own.

```
-- Gray code is additive over GF(2)
theorem sc_gray_gf2_linear :
  ∀ x y : Fin 32,
    scGray (x.val ^^^ y.val)
      = scGray x.val ^^^ scGray y.val
:= by native_decide
```

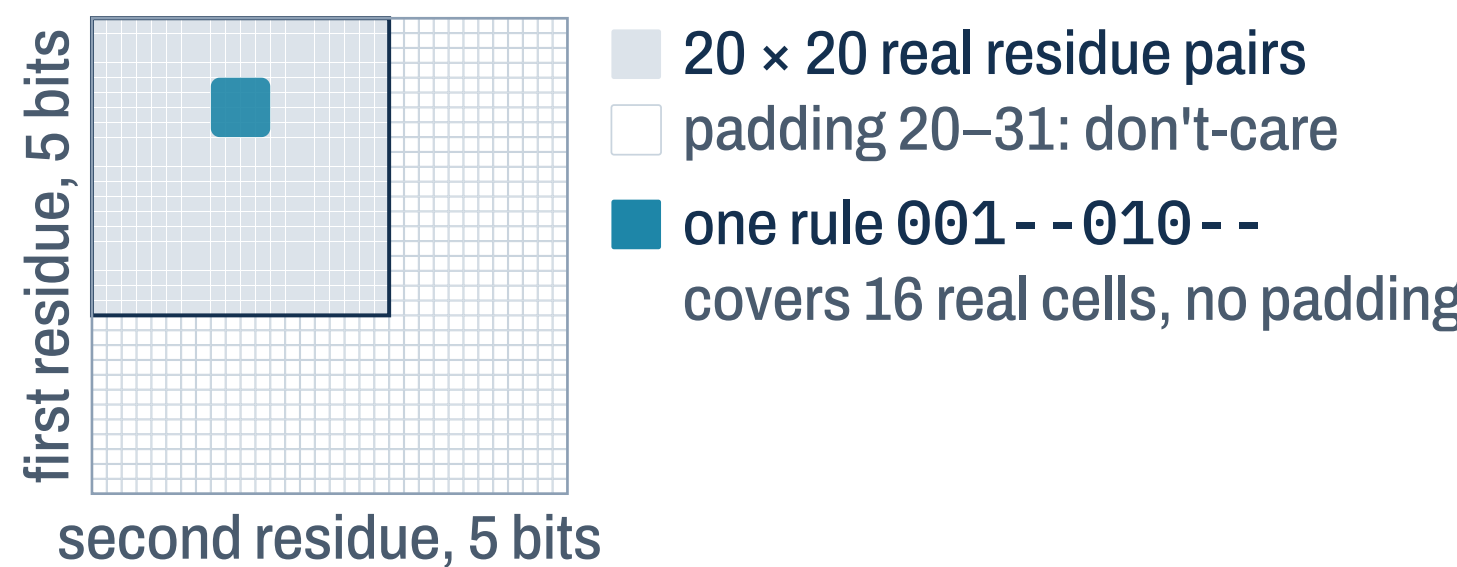
	native_decide	standard axioms only	no axiom
KmapProofs	16		
KmerIndexing	35		
AminoAcidEncoding	27		
ContactMapCompleteness	20		
KmapEncodingEquiv	8		
ContactCircuits	12		131 theorems
SequenceCircuits	13		100 + #23 + #8

101 What the proofs guarantee

Exact rules on padded maps

Twenty residues need five bits per axis, so a 20×20 residue-pair map is padded to 32×32 with 10-bit addresses. Lean proves that a fully fixed rule matches exactly its own cell and that such a rule on two real residues fires only inside the 20×20 block, never on padding.

$32 \times 32 = 1,024$ addresses (10 bits)

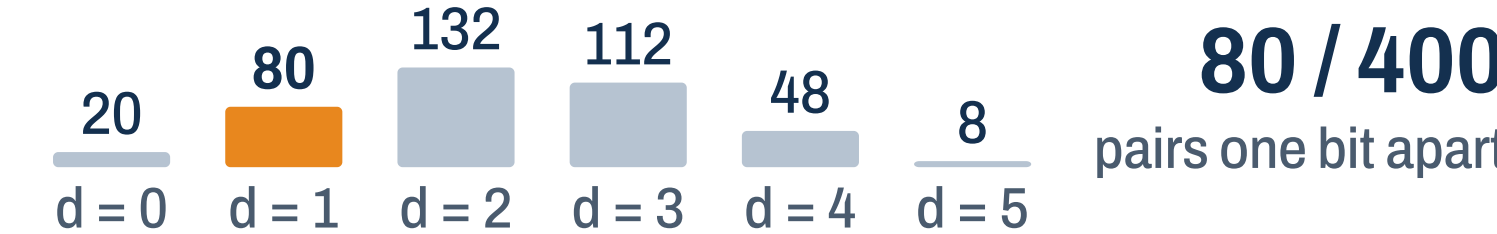


Same cube, still a segment, different residues

Plain binary	cube 10 - - $\rightarrow$ residues 8-11
00 00 00 00 01 01 01 01 10 10 10 10 11 11 11 11	10 10 10 10
00 01 10 11 00 01 10 11 00 01 10 11 00 01 10 11	00 01 10 11
01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01	01 01 01 01
10 10 10 10 10 10 10 10 10 10 10 10 10 10 10 10	10 10 10 10
Gray-coded	cube 10 - - $\rightarrow$ residues 12-15

Lexicographic (FCGR)	Gray	Same 16 counts, moved
A C G T	i $\rightarrow$ g(i)	A C T G
A 1 1 00 $\rightarrow$ 00 1 1		A 1 1 1
C 4 01 $\rightarrow$ 01 4 0		C 4 0 1
G 12 10 $\rightarrow$ 10 12 1		G 12 1 0
T 1 11 $\rightarrow$ 11 1 1		T 1 1 1
		G 0.976747 = 0.976747

What Gray changes: code distance d, 400 residue pairs



Crambin, 8,000 frames
a = contact 4-44, t = 4-43

	t formed	t broken
a formed	5,429	0
a broken	1,258	1,313

9,986

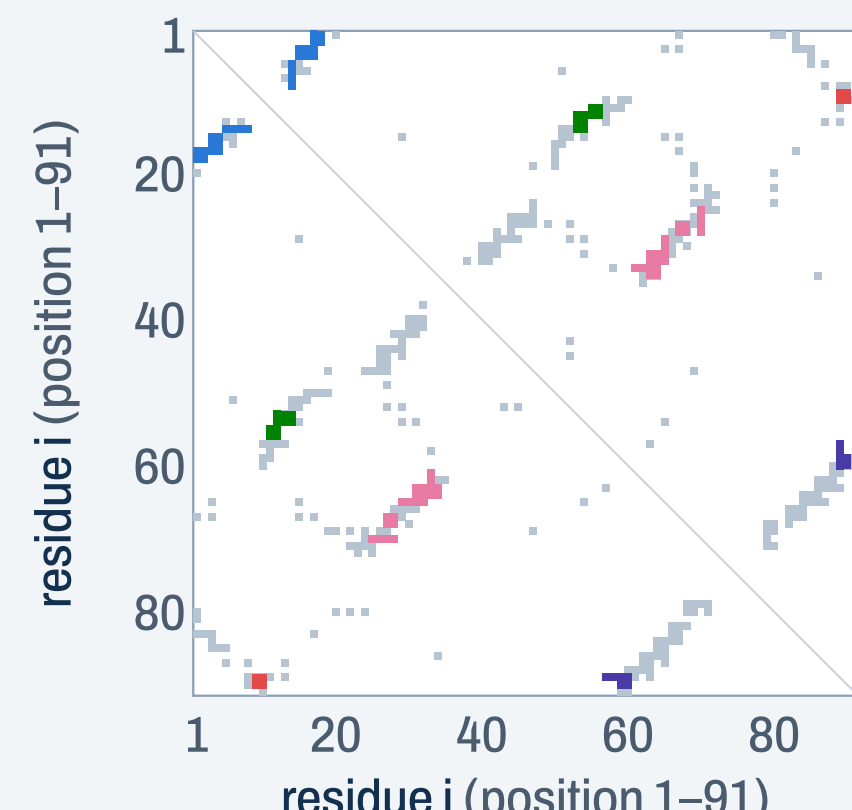
rules hold in every frame
of 479 folding runs
like $a \Rightarrow t$: a formed
never without t

3,966 join neighbouring contacts

33 / 33 in the long runs

100 Rules with no exception, from exact circuits

Each PSICOV⁷ protein (150 Pfam families) becomes a circuit equal to its contact map. Its AND gates are rules; block rules that continue one another give strand pairs, direction and register, by counting.



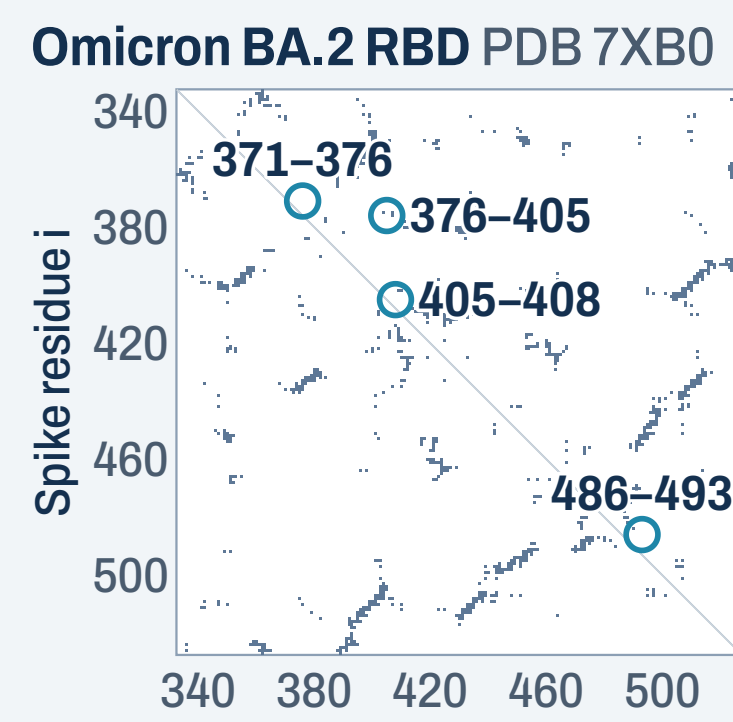
One AND gate, read as a rule
i 001101 - IF 27-28 AND 67-68
j 100001 - THEN contact (0 exceptions)

Read from the rules: strand pairs

- 25-34 $\cdot$ 61-70 antiparallel $i + j = 94-98$
- 1-8 $\cdot$ 14-18 antiparallel hairpin $i + j = 18-22$
- 11-14 $\cdot$ 53-56 antiparallel $i + j = 65-68$
- 57-60 $\cdot$ 89-90 parallel $j - i = 29-32$

150 / 150 proteins exact on all 6,905,856 inputs
27,997 rules, 0 exceptions, 0 contacts missed
378 antiparallel, 215 parallel pairs, 104 hairpins

One species, SARS-CoV-2 Omicron: in 1,299 Spike sequences, exact circuits list which residue pairs occur at co-evolving positions and which never do. No sequence breaks a rule.



Omicron BA.2 RBD PDB 7XB0

Rings: the 4 RBD rule pairs. None is a contact: $C\beta$ 12.8-17.5 $\text{\AA}$, or 3 apart (405-408).

371 $\times$ 376	405 $\times$ 408	486 $\times$ 493
1,294 sequences	1,289 sequences	1,297 sequences
A376 T376	R408 S408	Q493 R493
F371 970 0	D405 319 0	F486 20 1,173
L371 0 311	N405 12 958	P486 2 0
S371 1 12		V486 102 0

10 position pairs $\cdot$ 41 allowed and 35 forbidden residue combinations $\cdot$ 23 implications $\cdot$ 0 violations: every one of the 1,299 sequences obeys every rule

Takeaways

- A verified foundation. The encodings are Lean theorems; every circuit is exact on all inputs.
- Rules that are true, not likely. Each holds for every contact, sequence or frame it names.
- One language for sequence, structure and dynamics. Any finite encoding can be checked the same way.
- Lossless by proof. Each cell decodes to both residues; Gray order only permutes cells, so no statistic changes.
- Exact about absence. No rule on chemistry alone holds without exception: all 196 group $\times$ separation classes of PSICOV pairs mix contacts and non-contacts.

Demo, any PDB entry: e-motioner-x-sbs.github.io/ICMS_Submission

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* This work is currently under review.